WI Generator: Model Comparison

Part **265472208237** · SUPPORT LH/RH DOOR OUTER HANDLE · Stage 01 · Source: 265472208237 NEW PPAP.xlsx · Evaluated 10 November 2026

Summary

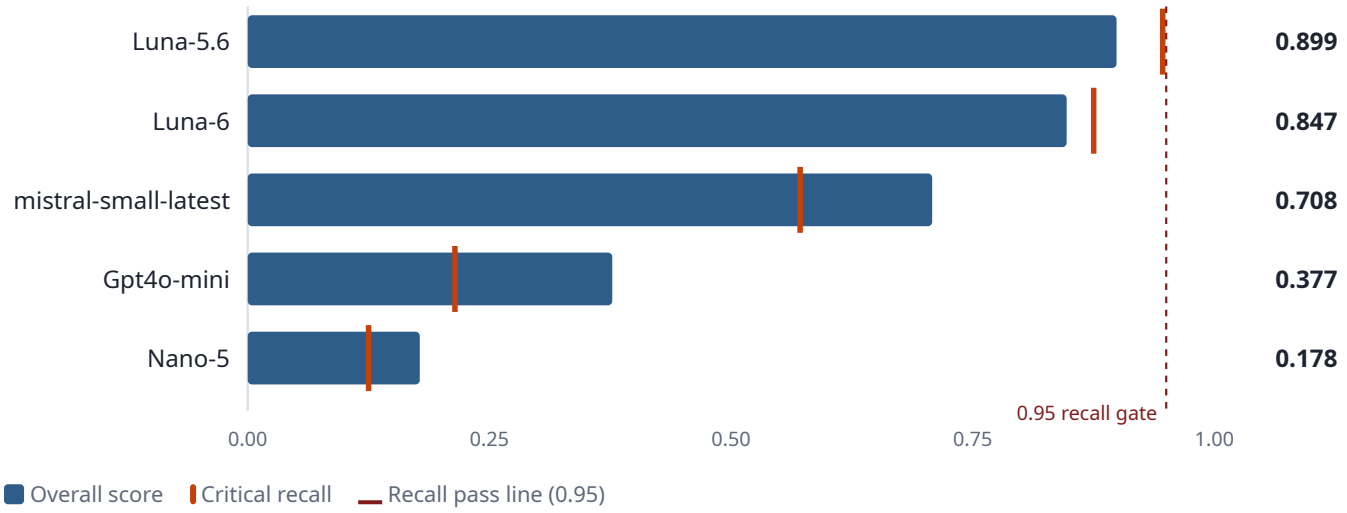
- **No model passes the judge.** Luna-5.6 comes closest: it has no contradicted critical fact, but it fails on 20 critical additions and on critical recall of 0.946, just under the 0.95 gate.
- **Luna-5.6 leads (0.899, recall 0.946), with Luna-6 second (0.847, recall 0.875).** Both write every operation (shearing, blanking 150 T, piercing 75 T, Forming I and II on 150 T) with the golden shut heights, strokes and dimensions on the right press. mistral-small has every press setting but no dimensions (0.708). gpt-4o-mini (0.377) and Nano-5 (0.178) do not produce a usable press process.
- **Luna-6 costs $0.004472 per WI**, about the same as gpt-4o-mini ($0.004213) and less than mistral-small ($0.005468). Luna-5.6 costs $0.008906, so its 0.052 lead costs 2.0×.
- **Almost all of the Luna models' critical additions are real PPAP data:** 19 of Luna-5.6's 20 and all 14 of Luna-6's. The golden leaves out control-plan rows the generator reads, including the 28 mm piercing dimension, R3 on Forming I and the rev-02 LH/RH bin segregation. Nano-5 copied the prompt's welding/fixture example almost word for word.
- **Some golden values cannot be derived from the PPAP:** the ±0.05 mm thickness tolerance, the plastic-mallet and rust-remover rework actions, and the “part missing → assemble all parts” rework row. If the tolerance were in the source, Luna-5.6's recall would be 0.964, above the gate.

Scorecard

#	Model	Overall	Critical recall	Critical contradicted	Critical hallucinations (in PPAP)	Language	Cost / WI	Result
1	**Luna-5.6** openai / gpt-5.6-luna	**0.899**	0.946	0	20 (19)	Compliant	$0.0089	FAIL
2	**Luna-6** openai / gpt-6-luna	**0.847**	0.875	1	14 (14)	Compliant	$0.0045	FAIL
3	**mistral-small-latest** mistral / mistral-small-latest	**0.708**	0.571	4	8 (4)	Compliant	$0.0055	FAIL
4	**Gpt4o-mini** openai / gpt-4o-mini	**0.377**	0.214	1	2 (2)	Non-compliant	$0.0042	FAIL
5	**Nano-5** openai / gpt-5-nano	**0.178**	0.125	2	6 (1)	Non-compliant	$0.0016	FAIL

Overall is the section-weighted score from 0 to 1. Critical recall is pooled over the stage. A model passes only with 0 contradicted critical facts, 0 critical hallucinations and critical recall of at least 0.95. Numbers in brackets are critical hallucinations whose content is present in the PPAP. Costs come from each run's own cost report.

Overall score vs. critical recall



Where each model scores

Cell colours: ≥0.90 0.60–0.89 0.30–0.59 <0.30

Section scores

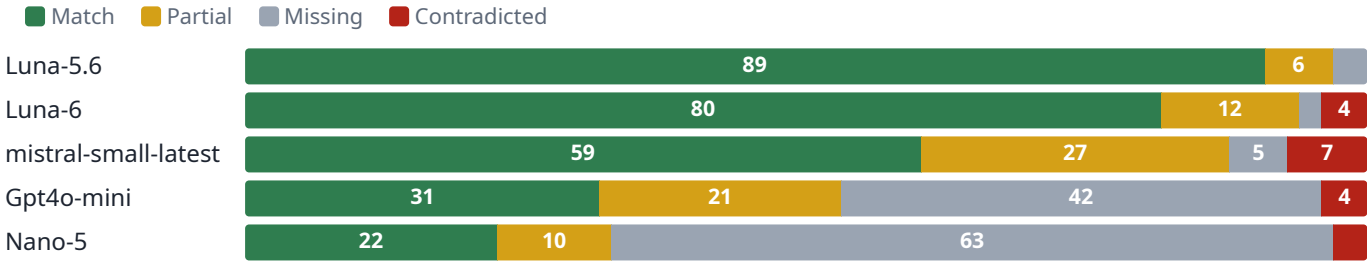
Section (weight)	Luna-5.6	Luna-6	mistral-small-latest	Gpt4o-mini	Nano-5
Steps (0.35)	0.94	0.96	0.68	0.26	0.01
Finished QC (0.15)	0.90	0.90	0.80	0.40	0.10
Rework (0.15)	0.67	0.42	0.25	0.17	0.33
Incoming QC (0.10)	0.85	0.85	0.85	0.75	0.00
Header (0.10)	0.99	0.79	0.76	0.58	0.60
Tooling (0.10)	1.00	1.00	1.00	0.18	0.00
Legend (0.05)	1.00	1.00	1.00	1.00	1.00
Overall	**0.899**	**0.847**	**0.708**	**0.377**	**0.178**

Weighted share of golden facts covered in each part of the process (steps, tooling and finished-QC facts grouped by the operation they belong to).

Coverage by process area

Process area	Luna-5.6	Luna-6	mistral-small-latest	Gpt4o-mini	Nano-5
Header & part list	99%	79%	76%	58%	60%
Incoming material	89%	89%	93%	70%	0%
Shearing	100%	100%	100%	41%	0%
Blanking (150 T)	78%	100%	44%	6%	0%
Piercing (75 T)	100%	100%	83%	25%	0%
Forming I (150 T)	100%	89%	67%	44%	0%
Forming II (150 T)	100%	100%	67%	0%	0%
Final inspection, storage, packing	94%	88%	75%	38%	12%
Inspection instruments	100%	100%	100%	18%	0%
Rework	67%	42%	25%	17%	33%
Legend	100%	100%	100%	100%	100%

Verdicts on the 98 golden facts



Findings by model

#1 Luna-5.6 openai / gpt-5.6-luna — Overall **0.899** — Crit. recall **0.946** — Facts matched **89/98** — $0.0089/WI

RIGHT The most complete WI: header 0.991, tooling 1.00, steps 0.942. Doc number, revision 02, dates, both part-list rows and every press setting and dimension match, including both 26 mm Forming I features. It has no contradicted critical fact.

WRONG

- The blanking chamfer 5×45° and the blanking R3 are missing (R3 appears only under Forming I and II), and the ±0.05 mm thickness tolerance is dropped. These are its only 3 critical facts that are not a MATCH.
- 20 critical additions, 19 of them in the PPAP: min/max shut heights, number of threads, tool and liner sizes for all four tools, hourly and first/last-piece checks, 28 mm, R3 on Forming I, LH/RH bins, RPO, UNIT-III transport and re-inspection, annual layout check, and four new rework rows.
- One addition is invented: radius R5 on Forming II (step 15 and QF 7). The PPAP lists only R3 and R4.5.
- Rework scores 0.667. Rusty raw material goes back to the supplier (row 1); only the finished-part row cleans and applies rust preventive. The radius/chamfer row readjusts but never replaces the tool, and there is no part-missing row.
- Line area written as “SHARADA INDUSTRIES / SUPPORT LH-RH DOOR OUTER HANDLE”, without UNIT-II.

#2 Luna-6 openai / gpt-6-luna

Overall **0.847** Crit. recall **0.875** Facts matched **80/98** $0.0045/WI

RIGHT Every operation is present with the golden shut height, stroke and dimensions on the right press, and the doc number, revision 02 and dates are correct. Steps score 0.958, the highest of the five, and tooling 1.00.

WRONG

- Stage No. written as “PP-010 to PP-110” instead of 1. This is its only contradicted critical fact. Luna-6 has written an operation-code range on all 7 stage-WIs scored so far, across four parts.
- The part list merges LH and RH into one row, “2654 7220 8237/8238N”, with quantity “As applicable (LH/RH)” (4 critical facts PARTIAL). Only one of the two 26 mm Forming I dimensions is listed.
- Rework scores 0.417. Rusty material is rejected and returned to the supplier instead of cleaned, the packing row corrects the logistic sheet instead of repacking, and there is no part-missing row.
- 14 critical additions, all of them in the PPAP: F-PRD-17, min/max shut heights, hourly, per-lot and first/last-piece sampling, 28 mm, R3 on Forming I, measuring tape, LH/RH bins, RPO, annual layout check, UNIT-III transport and two new rework rows.
- It used 3,813 of its 4,000 max_tokens. The output is complete, but a longer stage would be at risk of truncation.

#3 mistral-small-latest mistral / mistral-small-latest

Overall **0.708** Crit. recall **0.571** Facts matched **59/98** $0.0055/WI

RIGHT Every press setting is present and bound correctly: shut height and stroke for blanking, piercing, Forming I and Forming II, the shearing size 1250 × 165 mm, and all incoming checks. Tooling scores 1.00 and the text is in Hindi.

WRONG

- The chamfer 5×45° and R3 are “applied after cutting”, before blanking, instead of checked after blanking (2 contradicted critical facts).
- No blanking, piercing or forming dimension is given. Only “as per control plan” appears, so 20 dimension facts are PARTIAL or MISSING.
- Header: line area UNIT-3, stage name “R M RECEIPT”, machine field “Manual”, and part number with “/H/06/01/12” appended. The RH part-list row drops the N (2654 7220 8238).
- 8 critical additions: a 150 T shearing machine and a “sheet metal cutting machine” (neither in the PPAP), min/max shut heights, F-QAD/SI/RMQP, die-pad and pin tool items, LH/RH bins, and two new rework rows (adjust thickness, redo holes).
- Rework scores 0.25. Dimension faults are “adjusted as per control plan” with no tool action, and wrong forming is re-done on the part.

#4 Gpt4o-mini openai / gpt-4o-mini — Overall **0.377** — Crit. recall **0.214** — Facts matched **31/98** — $0.0042/WI

RIGHT Doc number, revision, dates, D513, incoming surface criteria and final inspection as per quality gate are correct. Its rust rework row (clean, apply anti-rust) matches the golden.

WRONG

- Blanking and Forming II are missing entirely. Piercing and Forming I have no shut height, stroke or dimension, and there is no 75 T press anywhere.
- Part number with "/H/06/01/12" appended (contradicted critical fact), line area UNIT-III, and an empty part list.
- Incoming checks are done after shearing, and piercing after Forming I.
- All body text is in English (language NON_COMPLIANT).
- Dimension faults are reworked on the part instead of adjusting or replacing the tool. It adds F-PRD-17 and a material-replacement rework row.

#5 Nano-5 openai / gpt-5-nano — Overall **0.178** — Crit. recall **0.125** — Facts matched **22/98** — $0.0016/WI

RIGHT Doc number, revision 02, part number, drawing number and part name are correct. Its part-missing rework row matches the golden, but only because both follow the assembly example.

WRONG

- The WI is a fixture-assembly procedure ("Collect all loose parts", "Clamp the fixture", unclamp and inspect) copied from the prompt's welding example. There is no shearing, blanking, piercing or forming, and no setting or dimension.
- Document date and revision date are 11/11/17 instead of 06.03.2026 (2 contradicted critical facts). Line area is UNIT-3, and the machine field holds the supplier code S-06220.
- 6 critical additions, 5 not in the PPAP: a fixture with a fixture number, a date-code card, and weld/attachment and misalignment rework rows.
- All body text is in English (language NON_COMPLIANT).

Root causes

1. The weaker models do not extract the press process from the PPAP

The P-C-P lists 11 operations (PP-010 to PP-110), and the golden covers the incoming checks, shearing, the four press operations, inspection, storage and packing. Only the Luna models carry all of it. The other three fail in different ways:

Model	What it has	What it lacks or replaces	Completion tokens
mistral-small	All press settings, shearing size, incoming checks	Every blanking, piercing and forming dimension	1,725 of 16,000
gpt-4o-mini	Outline of shearing, piercing, Forming I, without values	Blanking, Forming II, all settings and dimensions	608 of 4,000
Nano-5	Header only	The whole process; writes a fixture assembly instead	686 of 4,000

None of this is a token-limit effect: all three stopped far below their limits.

2. The prompt's filled-in example still leaks into the output

Nano-5 reproduces the welding example in `pcp_wi_generator/prompt_builder.py` almost verbatim: “Collect all loose parts and place them together”, “Clamp the fixture”, a fixture with a fixture number, a weld/attachment rework row, and the date 11/11/17. mistral-small writes “UNIT-3”, the example's line area; the PPAP says UNIT-II. (`prompt_builder.py` is not on this branch; the strings match the example documented in the runbook.)

11/11/17 is also the PFD form's revision date in this PPAP, so Nano-5's date could come from either source. The same leakage was reported on 289661208202 and 544283753302.

3. The golden leaves out control-plan rows, and the judge counts them as hallucinations

Critical additions found in the PPAP: **Luna-5.6 19 of 20, Luna-6 14 of 14**, mistral-small 4 of 8, gpt-4o-mini 2 of 2, Nano-5 1 of 6. The golden omits several P-C-P rows that the Luna models copy: the 28 mm piercing dimension, R3 on Forming I, M.Tape for shearing, min/max shut heights, threads, tool and liner sizes, F-PRD-17 and per-hour/per-lot sampling.

It also omits the change that created revision 02. The P-C-P change log for 06.03.2026 adds separate LH/RH bins with identification tags, and PP-070/PP-100 add RPO. The golden is at revision 02 but contains neither, so the models that include the rev-02 change are penalised for it.

4. Some golden values cannot be derived from the PPAP

The golden's **±0.05 mm** sheet-thickness tolerance does not appear anywhere in the PPAP; the P-C-P gives only 1.5. Every model scores this critical fact PARTIAL at best. For Luna-5.6 it is one of only three non-MATCH critical facts: crediting it would raise its recall from 0.946 to 0.964, above the gate.

“Plastic mallet” and “rust remover” are also absent. The golden's “part missing → assemble all parts and inspect” row does not fit a single pressed part; only Nano-5 matches it, by copying the assembly example. The PPAP reaction plans say “Sort / Reject”, “Sort / Reject & Sent back To Supplier” and “Resharpening / Replacement of stopper/Tool”. Luna-6 follows them for rusty material and is marked CONTRADICTED. Rework is the lowest-scoring section for both Luna models.

5. Header and part-list rules are not explicit enough

Revision: the golden uses the P-C-P's (Rev.No.) 02 and 06.03.2026, not the form's own Rev 0 / 11/11/2017. Four models got this right; Nano-5 took 11/11/17.

Part No.: gpt-4o-mini and mistral-small appended the change level from “Part Number / Latest Change Level: 2654 7220 8237/8238N/H/06/01/12”.

Stage No.: Luna-6 wrote “PP-010 to PP-110”. It has written an operation-code range on every one of the 7 stage-WIs scored so far (289661208202, the four stages of 290058600101, 544283753302 and this part).

Part list: for this LH/RH pair the golden has one row per part number with quantity 1. Luna-6 and Nano-5 merged them into one row, and mistral-small dropped the N from 8238N.

Recommendations

1. **Replace the filled-in welding example in `prompt_builder.py`** with placeholders or a press-shop example, and tell the model never to copy example values. It leaks here (Nano-5 verbatim; mistral-small's UNIT-3), as it did on 289661208202 and 544283753302.
2. **Add explicit header rules to the prompt:** Stage No. is the stage number, never an operation-code range (Luna-6 has got this wrong on all 7 stage-WIs). Part No. is written without the change level. Revision and date come from the control plan's (Rev.No.) / Date (Rev.) fields, not the form's own revision. An LH/RH pair gets one part-list row per part number, each with its quantity.

3. **Tell the generator to give every dimension listed for each operation**, not “as per control plan”. mistral-small has every setting but no dimensions.
4. **Review the golden with its author:** the source of the ±0.05 mm tolerance, the plastic-mallet, rust-remover and part-missing rework rows, and whether the rev-02 LH/RH bin segregation, RPO, the 28 mm piercing dimension and R3 on Forming I should be in the WI. As it stands, the golden rewards copying the example and penalises reading the PPAP.
5. **Add a source-supported class to the judge** (e.g. `SOURCE_SUPPORTED_ADDITION`) for extra content traceable to the PPAP, so that it does not trip the hallucination gate. On this part, that would remove 19 of Luna-5.6's 20 critical additions and all 14 of Luna-6's.
6. **Keep testing Luna-5.6 and Luna-6 only.** Luna-5.6 has now led on 6 of the 7 stage-WIs scored (this part, 544283753302 and the four stages of 290058600101), and Luna-6 on one (289661208202), at half the cost. Raise max_tokens above 4,000 for both: Luna-6 used 3,813 here. Decide after re-running with the fixed prompt; this report covers one WI per model.

Method

Judge. Each generated WI was compared with its golden WI (`Hindi/WI_20260616_265472208237 NEW PPAP_Stage_01.xlsx`) using `wi_judge_single_prompt.md`. Claude applied the prompt; `eval_tools/score.py` computed every score from the verdicts, so the arithmetic is exact and auditable.

Golden facts. Each golden WI was split once into atomic facts, and the same list was used for every model. Each fact is MATCH (1.0), PARTIAL (0.5), MISSING (0) or CONTRADICTED (0), weighted 3 / 2 / 1 for critical / major / minor.

	Facts	Critical	Major	Minor	S	QF	R	QI	H	T	L
Section weight					0.35	0.15	0.15	0.10	0.10	0.10	0.05
Stage 01	98	56	25	17	42	5	6	4	24	8	9

Pass rule. A model fails if any critical fact is contradicted, if there is any critical hallucination, or if critical recall is below 0.95.

Judgement calls applied to all models:

- Forming I lists 26 mm twice and Forming II lists 25 mm twice. These are scored as separate facts because the P-C-P has them as separate characteristics. Luna-6 lists 26 mm once, so one of them is MISSING.
- A merged LH/RH part-list row (Luna-6, Nano-5) is PARTIAL for every part-list fact. Luna-6's quantity “As applicable (LH/RH)” is PARTIAL (field present without a number) rather than CONTRADICTED.
- mistral-small lists its machines only in the tooling list, and its header machine field says “Manual”. The header machine-list facts are PARTIAL; the operation→machine bindings are MATCH via the tooling list.
- The UNIT-II → UNIT-III transport (Luna-5.6, Luna-6) is a new process step and listed as a critical addition, not as the golden's dispatch. Luna-6 packs but never dispatches, so its pack-and-dispatch fact is PARTIAL.
- The golden's “check the lab report” step is PARTIAL when the lab report is not named (both Luna models), matching the rule used on 544283753302.
- Rejecting or replacing a dented part is PARTIAL, not CONTRADICTED, because the golden's action is “fix with plastic mallet *or replace the part*”.
- mistral-small's “apply chamfer 5×45° and R3 after cutting” is CONTRADICTED: the values are bound to shearing instead of to the blanking check.

Limits. This covers one part, one stage and one generation per model. Treat the ranking as indicative until it is repeated on more WIs.

Detailed results (per-fact verdicts, evidence quotes, extra content, score workings):

- `Generated/Luna-5.6/judge_result_Luna-5.6_Stage_01.json`
- `Generated/Luna-6/judge_result_Luna-6_Stage_01.json`
- `Generated/mistral-small-latest/judge_result_mistral-small-latest_Stage_01.json`
- `Generated/Gpt4o-mini/judge_result_Gpt4o-mini_Stage_01.json`
- `Generated/Nano-5/judge_result_Nano-5_Stage_01.json`